From Fast Prediction to Defensible Action: A Multi-Subhypothesis Research Report on Trustworthy Transient Stability Assessment for Renewable-Rich Power Systems

Research Report Generated from the Supplied Literature Corpus
Literature synthesis and hypothesis-design report
Power-system transient stability, trustworthy machine learning, and safe control

***Abstract*—This report synthesizes 45 supplied papers on transient stability assessment (TSA) for renewable-energy-rich power systems. The corpus contains strong but largely parallel advances in fast data-driven assessment, uncertainty and credibility estimation, Lyapunov/barrier certificates, converter-dominated dynamics, and transfer or active learning. We formulate six complementary subhypotheses (SHs), review their evidence roles, and identify both within-SH and cross-SH gaps. The central synthesis is that present pipelines rarely close the loop from a prediction's epistemic credibility, through a mode-dependent physical stability boundary, to an operationally feasible safety action. Two falsifiable hypotheses are proposed: (H1) a credibility-calibrated hybrid stability certificate can reduce false-safe decisions at equal operational cost; and (H2) certificate-violation-risk-driven active transfer can reduce electromagnetic-transient (EMT) labeling cost while preserving safety under operating drift. A staged simulation, hardware-in-the-loop, and ablation plan is provided. The proposed work deliberately distinguishes corpus coverage, compatible direct evidence, and scientific conclusion; the hypotheses are research propositions to be tested, not claims already established by the literature.**

***Index Terms*—transient stability assessment, renewable energy, inverter-based resources, trustworthy machine learning, control barrier function, Lyapunov certificate, transfer learning, safe reinforcement learning**

I. Research Question and Scope

Power systems with large shares of inverter-based resources (IBRs) require decisions faster than repeated high-fidelity time-domain or EMT simulation can usually provide. The supplied corpus addresses this need from several directions: supervised TSA and stability-margin prediction, probabilistic risk estimation, interpretable or credible learning, direct-method certificates, converter synchronization analysis, and adaptive learning. The broad research question is therefore:

> *How can a real-time TSA system turn uncertain, distribution-shifted trajectory information into a mode-aware and operationally safe decision without treating a prediction score as a physical certificate?*

This report is a structured literature-derived research design rather than a systematic review claiming exhaustive external coverage. The 45 PDFs in the supplied corpus were screened by title, abstract, and selected method/limitation sections. Evidence is organized into six SHs whose functions are intentionally different; they are not six repetitions of a single "baseline–adverse–boundary" template.

II. Iterative Synthesis Process

The synthesis followed five iterations. First, the corpus was grouped by the scientific function of each paper rather than by a single causal chain: prediction, uncertainty/credibility, stability geometry, physical mode, control, and adaptation. Second, candidate directions that merely restated a published method were removed. Third, each remaining direction was expressed as a claim that can be evaluated by compatible evidence: simulation and calibration metrics for predictors, Lyapunov/barrier conditions for certificates, and EMT or hardware-in-the-loop (HIL) tests for converter dynamics. Fourth, interdependencies were mapped. Fifth, only gaps that have an identifiable test, comparator, and falsification condition were retained.

The resulting logic is

$$\begin{aligned}&\text{trajectory and operating state} \rightarrow \text{prediction and credibility}\\&\rightarrow \text{mode-aware certificate} \rightarrow \text{safe action and feedback.}\end{aligned} \tag{1}$$

where every arrow has a separately testable failure mode. This avoids the common mistake of asking one article to establish prediction accuracy, physical validity, generalization, and closed-loop safety all at once.

III. Multi-Subhypothesis Literature Review and Local Gaps

Table I summarizes the six SHs. "Direct evidence" below means evidence directly compatible with the stated SH, not necessarily a controlled intervention experiment. For example, a formal Lyapunov condition is direct theoretical evidence for an invariant-set claim, whereas EMT trajectories are direct empirical evidence for a converter-transient claim.

A. SH1: Fast Assessment Is Not Yet a Decision Guarantee

The literature shows mature progress in making TSA fast and uncertainty-aware. Lu and Bu predict probabilistic post-fault rotor-angle trajectories and make their temporal distribution visible to operators [1]. Tan and Zhao address the

TABLE I

SIX COMPLEMENTARY SUBHYPOTHESES, CURRENT EVIDENCE, AND USABLE GAPS

SH	Research function and literature signal	Current limitation	Usable gap and direct test
SH1	Fast TSA, margins, and probabilistic risk. Graphical learning predicts probabilistic rotor-angle trajectories [1]; debiased uncertainty targets imbalanced instability data [2]; DSPP supports chance-constrained preventive control [3].	A confidence interval or probability is usually consumed as a score, not converted into a state-dependent operational safety margin. Calibration under rare, shifted events is not equivalent to safe action.	Learn a calibrated risk envelope that explicitly penalizes false-safe decisions. Test coverage, Brier/ECE, false-safe rate, and curtailment cost under unseen faults, topologies, and renewable scenarios.
SH2	Credibility, OOD detection, and error anticipation. Hybrid credibility learning combines ensemble consistency and data-disparity indicators [4]; per-sample misclassification prediction supplies a complementary error signal [5].	Existing credibility methods can abstain or warn, but generally do not identify which physical boundary or controller constraint should become tighter.	Map credibility to a measurable uncertainty inflation and a safe-set contraction. Test whether credibility-conditioned actions dominate fixed thresholds at the same reliability target.
SH3	Physical stability certificates and safe control. Neural Lyapunov control learns a controller with a falsification module [6]; barrier-certified RL filters learned actions to meet predefined safety regions [7].	A nominal certificate can be invalidated by model mismatch, distribution shift, switching, and unmodeled converter current dynamics.	Build certificates conditioned on identified instability mode and uncertainty radius. Verify barrier/Lyapunov residuals and worst-case post-fault trajectories, not classification accuracy alone.
SH4	Hybrid and converter-dominated stability geometry. Multiple constrained stability regions model renewable trips [8]; current transients materially affect PLL synchronization assessment [9]; multi-VSC interaction changes dominant synchronization risk [10]; unstable limit cycles can form the relevant boundary [11].	A binary rotor-angle TSA label masks distinct rotor, voltage, PLL, protection/trip, and limit-cycle mechanisms. A single boundary may be physically misleading.	Infer a dominant instability mode and invoke a mode-specific certificate. Test mode accuracy and boundary-violation detection on switching, LVRT/FRT, PLL, and multi-VSC scenarios.
SH5	Transfer, active learning, and data efficiency. Bidirectional active transfer lowers the demand for labels across conditions [12]; temporal transfer addresses temporal feature mismatch [13]; physics-augmented auxiliary learning improves representation and margin calibration [14].	Label selection optimizes representativeness or classifier adaptation, not the decision-critical risk of violating a safety certificate after deployment.	Use OOD, predicted error, and certificate residual to choose EMT labels. Test labels-to-safety efficiency against random, entropy, and OOD-only sampling.
SH6	Operational integration. Chance-constrained preventive control provides a link between uncertainty and set-points [3]; safe RL offers a control filter [7].	Assessment, certification, control, and post-event model update remain modular; no clear contract determines when an uncertain forecast must trigger conservative control, further simulation, or abstention.	Define a three-action policy: act, constrain, or request high-fidelity evaluation. Test closed-loop security, recovery, and operational cost.

imbalance and multimodality that make uncertainty estimates unreliable exactly where rare instability matters [2]. Su *et al.* place a Bayesian surrogate inside chance-constrained preventive control [3]. These are essential ingredients, but they leave a decision gap: the threshold converting a statistical output into an action is often fixed, although prediction error and operating consequences are state-dependent.

The SH1 gap is therefore not "improve accuracy" in the abstract. It is to estimate a conservative decision risk $\mathcal{R}(x)$ whose calibration is evaluated especially on the false-safe tail. A negative result is informative: if a calibrated risk envelope cannot outperform a single threshold at matched curtailment cost, the additional machinery does not justify operational use.

B. SH2: Credibility Must Be Actionable

Credibility research correctly recognizes that high in-distribution accuracy does not establish reliability under changing conditions. Hybrid credibility learning uses ensemble behavior, outlier degree, and one-sample MMD, including misclassified OOD examples in training [4]. Misclassification prediction supplies a per-sample probability that an otherwise fast TSA classifier is wrong [5]. However, a warning has no intrinsic operational meaning. The system must decide whether to keep the nominal set-point, impose a conservative constraint, or obtain more authoritative evidence.

The SH2 gap is an explicit calibration-to-action contract. Let $c(x) \in [0,1]$ be credibility and $u(x)$ a calibrated uncertainty radius. The report proposes that $u(x)$ changes the admissible safety set rather than merely coloring a dashboard:

$$\mathcal{S}_{\mathrm{rob}}(x) = \{a : \max_{\delta \in \Delta(u(x))} B_{m(x)}(f(x,a,\delta)) \geq 0\}, \quad (2)$$

where $B_{m(x)}$ is a mode-specific barrier function and Δ expands as credibility decreases. This equation is a testable design proposal, not a claim that the available credibility models already ensure safety.

C. SH3: Certificates Are Strong but Need Deployment Validity

Lyapunov and barrier methods contribute a qualitatively different kind of evidence from a predictor: under their stated assumptions, they establish invariance or decrease conditions. Neural Lyapunov control jointly searches for a control law and a candidate Lyapunov function, with falsification used to enforce conditions [6]. Barrier-certified RL similarly filters learned actions against a neural barrier certificate [7]. Their limitation in the combined setting is external validity: a certificate for a nominal model does not necessarily remain valid when input trajectories are shifted or when the relevant physical mode has changed.

The local gap is a certificate that declares its domain of validity. It should report the physical mode, uncertainty radius, residual violation statistics, and whether it was verified by RMS simulation, EMT simulation, or HIL. This separates theoretical validity from empirical support rather than combining them into one opaque score.

D. SH4: The Stability Boundary Is Hybrid and Mode-Dependent

The physical-literature branch warns against treating all insecurity as the same rotor-angle classification problem. Mishra *et al.* model prone-to-trip renewable-rich systems as switched systems with multiple LVRT-constrained stability regions [8]. Chen *et al.* show that neglecting converter current transients can cause inaccurate synchronization assessment [9]. Wang *et al.* further show that nonlinear damping and transient interaction among VSCs determine dominant-converter risk [10]. Xu *et al.* explain that an unstable limit cycle, not only an unstable equilibrium point, may shape the relevant VSC boundary [11].

The usable SH4 gap is a *hybrid boundary atlas*: a mode classifier maps a trajectory to rotor-angle, voltage, PLL-synchronization, protection/trip, or limit-cycle-dominated regimes, and each regime uses an appropriate stability function. A rigorous failure case would be mode confusion that increases the false-safe rate relative to a single robust classifier; this must be measured, not assumed away.

E. SH5: Adaptation Should Spend Labels Where Safety Is Most Fragile

Transfer and active learning reduce the cost of adapting to operating changes. Bidirectional active transfer uses both useful target instances and selective removal of source instances [12]; temporal transfer specifically aligns evolving time-series features, including zero-target-sample use cases [13]. Physics-augmented auxiliary learning offers a mechanism-guided route to margin prediction and generalization [14]. Yet a label selected because it is uncertain or representative may be operationally unimportant, while a rare trajectory close to a certificate violation can be decisive.

The SH5 gap is a safety-value acquisition function. Instead of selecting only by entropy, choose the event whose high-fidelity label is expected to most reduce robust-boundary violation. The key result is not average accuracy at a fixed label budget, but the area under a labels-versus-false-safe curve.

F. SH6: An Operational Contract Is Still Missing

The existing pieces suggest, but do not complete, an online operating loop. A credible predictor may flag OOD; a physical certificate may filter an action; a preventive optimizer may adjust set-points. The missing contract is what happens when these disagree. In particular, a low-confidence but nominally stable prediction should not automatically be labeled safe, and a conservative certificate need not force unnecessary curtailment if a rapid EMT/HIL check is available.

The SH6 gap is a policy with three outcomes: *act* when risk is low and certification holds; *constrain* when risk is elevated but a robust action exists; and *escalate* when credibility or certificate residual is inadequate. This turns model uncertainty into an auditable engineering workflow.

IV. Detailed Gaps for Converter-Dominated, Renewable-Rich Grids

The following refinement makes the gap statements operational for high-renewable-penetration grids with power-electronic equipment. The quantitative values in this section are *reported evidence anchors* from the supplied papers, not numbers that are assumed to transfer unchanged across networks, fault-clearing practices, or controller settings. Cross-paper accuracy is not treated as a leaderboard because the fault sets, class balance, PMU windows, and security labels differ.

A. SH1: Risk and Margin Must Retain Physical Units

What is established. Fast predictors are now sufficiently capable that their output can be used inside an online loop. A hierarchical PMU-based model reported an average response time of 16 ms over 3000 test cases, a stability-margin-prediction error below 0.036, and accuracy above 98.5% on its reported unknown scenarios [20]. An improved deep Lyapunov learner reported 98.95% overall accuracy and 99.28% precision on its test case [15]. For probabilistic assessment at scale, PPGP was evaluated on a Texas 2000-bus system with 471 PV and 470 wind generations [16]; a separate large-scale EMT framework reported approximately 400-fold acceleration for a synthetic IEEE 118-bus AC/DC system containing two million RES entities [17]. These results establish computational feasibility, not a universal safety guarantee.

Precise gap. The common output is a class probability, a transient stability index, a critical clearing time (CCT), or a scalar margin. Such values are often not tied to the *physical limiting event*: first-swing angle separation, voltage recovery failure, PLL loss of synchronism, current saturation, LVRT/FRT tripping, or a controller state reaching its hard limit. Consequently, a well-calibrated probability can still be false-safe if the latent model does not represent the active boundary.

Researchable SH1 claim. For each contingency e, estimate a vector

$$\mathbf{y}_e = [\widehat{\mathrm{CCT}}, \widehat{\Delta\delta}_{\max}, \widehat{\Delta V}_{\min}, \widehat{\omega}_{\mathrm{PLL}}, \widehat{I}_{\max}, \widehat{p}_{\mathrm{trip}}], \tag{3}$$

along with prediction intervals. The margin used for action is the minimum normalized distance to the applicable protection, synchronization, voltage, and thermal/current constraints, rather than an unqualified TSA label. CCT estimation work already treats the CCT as a stability-boundary quantity [18]; trajectory-sensitivity work shows that sensitivity norms can rank control and reference parameters that move the DFIG-WTG synchronization boundary [19].

Quantitative test and falsifier. Stratify test events by CCT distance, IBR penetration, short-circuit ratio, fault duration/location, and control mode. Report root-mean-square error for every physical component in (3), 90% interval coverage, expected calibration error (ECE), and false-safe rate within the most critical 10% of the CCT/margin distribution. A planned success criterion is a 95% bootstrap upper confidence bound on false-safe rate below 1% in the critical stratum; failure of coverage or a false-safe concentration near a limiter invalidates the proposed margin mapping even if global accuracy remains high.

B. SH2: Credibility Requires a Decomposition, Not One Threshold

What is established. Credibility can be learned from model consistency and distributional disparity [4]; scene-dependent credibility has also been formulated using a localized generalization-error bound and a stability-density construction. In its reported cases, a critical scene-credibility index of 0.93 separated results with 100% stated accuracy [21]. These methods are valuable alarms, but their scalar score does not distinguish sensor corruption, operating-point extrapolation, topology change, label ambiguity near a boundary, or absent converter physics.

Precise gap. A single credibility score can conceal materially different operational responses. PMU noise or a missing feature should request data-quality checking; a novel grid strength or PLL parameter should enlarge model uncertainty; an imminent hard-limit encounter should immediately shrink the feasible control set. In addition, a credibility classifier trained on rotor-angle labels may look confident while the actual event is short-term voltage instability or converter synchronization loss.

Researchable SH2 claim. Replace $c(x)$ by

$$c(x) = [c_{\text{data}}, c_{\text{shift}}, c_{\text{mode}}, c_{\text{model}}, c_{\text{certificate}}], \qquad (4)$$

where the entries respectively measure data integrity, OOD distance, uncertainty in dominant-instability-mode (DIM) assignment, predictive disagreement, and certificate residual. The policy must state which component triggered escalation. This is supported by evidence that a networked shapelet TSA retained over 98% reported accuracy under studied cases and lost less than 0.7 percentage points under network-parameter errors, while its PMU-noise tests explicitly ranged from 0.5% to 3% [22]; that robustness observation should not be extrapolated to unmodelled IBR control changes.

Quantitative test and falsifier. Construct separated stress suites: PMU noise/dropout, line/topology change, grid-strength change, renewable forecast error, and PLL/current-control mismatch. Compare one scalar threshold with the vector policy in (4) at equal intervention cost. Report per-source AUROC for error detection, ECE, mode-conditional coverage, and the percentage of escalations that are later verified necessary by EMT. The decomposition fails if it raises the total escalation rate without improving calibration, false-safe rate, or attribution correctness.

C. SH3: Certificate Validity Must Include Saturation and Converter Timescales

What is established. Neural Lyapunov and barrier methods provide a formal safety mechanism under their stated state-space and model assumptions [6], [7]. Deep Lyapunov learning further links the learned decision function to a region-of-attraction construction [15]. Yet a power-electronic grid is a differential-algebraic, switched and constrained system: current controllers, PLLs, DC-link/AC voltage dynamics, virtual inertia, ride-through logic, protection, and anti-windup limiters introduce mode switches and time-scale separation.

Precise gap. A certificate $V(x)$ or $B(x)$ learned from nominal RMS trajectories may remain positive while the real control action enters current limitation or a PLL transient invalidates the reduced model. The hard-limit formulation of Romay *et al.* shows that control bounds must be enforced inside the dynamic equations through complementarity conditions, rather than checked after a trajectory is generated [23]. This exposes a missing contract between learned certificates and constraint-active dynamics.

Researchable SH3 claim. Use a hybrid certificate bank $\{B_m, V_m\}$ indexed by both DIM and limiter state m:

$$\dot{V}_m(x, a) \leq -\alpha_m \|x - x_m^\star\|^2 + \rho_m u(x), \qquad B_m(x) \geq 0, \quad (5)$$

where $u(x)$ is the credibility-derived uncertainty radius. Mode transitions must satisfy reset or dwell-time conditions, and each certificate must declare whether its evidence comes from analytical residual checks, RMS simulation, EMT simulation, or HIL.

Quantitative test and falsifier. Record the maximum violation of (5), time spent in current/voltage saturation, PLL frequency/angle excursion, and violation rate after a mode switch. Compare a nominal smooth certificate, a hard-limit-aware certificate, and the hybrid bank. The SH is not supported if robust filtering avoids violations only by excessive load shedding/curtailment, or if certificate residuals cease to predict EMT violations under current transients.

D. SH4: Dominant Instability Modes Need a Boundary Atlas

What is established. High-renewable systems can exhibit coupled but nonidentical rotor-angle, short-term voltage, and converter-synchronization phenomena. DIM work explicitly targets the distinction between rotor-angle and short-term voltage instability. A local-global GNN/Transformer formulation found its peak reported accuracy with a 2 s window and training time below 3 s per epoch, and illustrated that DIM-guided control selects generator shedding versus load shedding differently [24]. The multi-VSC study includes nonlinear damping, angular-frequency jumps, and transient interactions, with PSCAD/EMTDC and RT-LAB validation [10]. PLL current transients cannot be neglected without distorting synchronization assessment [9]. Finally, an unstable limit cycle can be the relevant VSC stability boundary rather than an unstable equilibrium point [11].

Precise gap. Current DIM methods mostly assign a label, while direct methods mostly analyze one specified mechanism.

The missing object is a *boundary atlas* that maps a trajectory to (i) a dominant mechanism, (ii) the correct reduced state and observable, (iii) a validity domain, and (iv) an appropriate remedial action. Without it, a rotor-angle surrogate may be asked to judge an LVRT trip cascade, and a voltage-oriented action may be applied to a PLL-dominated event.

Researchable SH4 claim. Let $m \in \{\mathrm{RA}, \mathrm{STV}, \mathrm{PLL}, \mathrm{trip}, \mathrm{ULC}\}$ denote rotor-angle, short-term voltage, PLL synchronization, protection/trip, and unstable-limit-cycle mechanisms. Learn $p(m \mid z)$ from PMU and converter features, but use mode-specific observables and action sets: rotor-angle separation/CCT; voltage nadir and recovery time; PLL angle-frequency/current excursion; LVRT/FRT dwell/trip count; and center-manifold/limit-cycle distance, respectively. The confidence of this assignment becomes the c_{mode} term in (4).

Quantitative test and falsifier. Evaluate macro-F1 and a cost-weighted confusion matrix, not only overall DIM accuracy. Every misclassification must be labeled as benign, conservative, or false-safe after EMT replay. Required scenario axes include IBR share, short-circuit ratio, number of interacting VSCs, fault impedance/duration, PLL bandwidth, current limit, LVRT/FRT setting, and protection delay. A conservative planning target is to reduce false-safe cross-mode errors by at least 30% relative to a binary TSA baseline at matched remedial-action energy; the atlas is rejected if extra modes merely redistribute errors without lowering the false-safe cross-mode subset.

E. SH5: Data Adaptation Should Target Certificate Failure, Not Only Classifier Uncertainty

What is established. Multiple techniques reduce simulator and labeling cost. Transfer ELM reported more than 24% improvement in stability-assessment accuracy in its large-scale cross-scale transfer setting [27]. Instability-pattern-guided updating began from 98.37% test accuracy with 181 errors and used a 556-sample representative set, about 5% of its original dataset, to constrain update cost [25]. A topology-clustering/DCGAN study reported, for one scenario, accuracy/security/recall of 97.40/97.81/88.10%, compared with 91.64/92.14/80.13% for random oversampling, but explicitly notes that its target is global transient power-angle stability rather than other instability modes [26]. These results show that sample efficiency is possible, while also revealing the class and mode scope of current targets.

Precise gap. Entropy, margin, or OOD sampling may spend an EMT label on a novel but operationally harmless trajectory, while undersampling a familiar-looking event near a hard current limit or mode transition. Updating a classifier on a pattern can also cause certificate drift or catastrophic forgetting in regions that were previously safe.

Researchable SH5 claim. Schedule high-fidelity labels with

$$\begin{aligned} A(z) = {} & \alpha d_{\mathrm{OOD}}(z) + \beta \widehat{p}_{\mathrm{err}}(z) \\ & + \gamma r_{\mathrm{cert}}(z) + \eta H[p(m \mid z)] \\ & + \kappa L_{\mathrm{impact}}(z), \end{aligned} \tag{6}$$

where r_{cert} is a barrier/Lyapunov residual and L_{impact} estimates load loss, curtailment, or protection consequence. The update has a replay/constraint term that preserves previously verified certificate margins.

Quantitative test and falsifier. Build learning curves of EMT labels versus false-safe rate, mode-macro-F1, ECE, certificate violations, and retained performance on prior scenarios. Compare random, entropy, margin, OOD-only, and (6) under identical EMT budgets. A pre-registered target is at least 25% fewer new EMT labels to reach the baseline's false-safe rate, with no greater than one percentage-point loss on the source-regime safety score. Failure on either the safety or forgetting condition rejects the active-transfer claim.

F. SH6: Control Must Optimize Security, Recovery, and Feasibility Together

What is established. Control-oriented TSA is no longer only conceptual. A master-slave TSC-OPF framework reports millisecond real-time dispatch on New England and Australian systems while incorporating contingency-specific surrogate feedback [28]. An integrated TSA-monitoring approach uses spatial-temporal features and critical-generator identification to select generator tripping for impending instability [29]. For converter control, a multi-VSG plus resistive superconducting fault-current-limiter study reported 62.12% current limiting during fault, 41.24% after clearing, a maximum 3.3% power vacancy, and 44.59% less power vacancy than its DDPG comparator in the reported cases [30]. These are valuable proof points, but they optimize different operational objectives under different mechanisms.

Precise gap. Existing control integrations generally assume a correct TSA/DIM result and do not expose what the controller should do when credibility is low, the mode is ambiguous, or a certificate is invalid. They also rarely compare generator tripping, load shedding, reactive support, VSG virtual-inertia/damping set-points, current-limit management, and topology actions on a common false-safe/cost scale.

Researchable SH6 claim. Solve a receding-horizon, three-action decision:

$$\begin{aligned} \min_{a \in \mathcal{A}_m} J(a) = {} & w_s \widehat{p}_{\mathrm{FS}}(a) + w_e E_{\mathrm{shed}}(a) \\ & + w_r T_{\mathrm{recovery}}(a) + w_c C_{\mathrm{wear}}(a), \end{aligned} \tag{7}$$

subject to the appropriate hybrid certificate, equipment limits, and a latency budget. If no feasible robust action exists, escalate to EMT/HIL or a predefined protection policy. The objectives make a false-safe declaration more costly than a conservative action, while preventing a nominally safe controller from achieving its score through indiscriminate curtailment.

Quantitative test and falsifier. Report delivered load/energy, renewable curtailment, generator/VSC trips, peak fault current, voltage recovery time, PLL resynchronization time, CCT improvement, and end-to-end decision latency. Compare with fixed rules, TSA-only remedial action, chance-constrained OPF, and nominal barrier RL. The control integration is unsuccessful if it cannot satisfy the selected protection/ride-through constraints within the latency budget, or if equal-security comparisons show no benefit beyond additional shedding.

V. Existing Research Limitations at the Collection Level

The six SHs expose four collection-level limitations.

1) **False-safe asymmetry is underrepresented.** Average accuracy, even with standard uncertainty intervals, does not reflect the much higher cost of incorrectly declaring a post-fault state safe.
2) **Epistemic and physical uncertainty are decoupled.** OOD or misclassification indicators identify when a predictor may be unreliable, but are not generally propagated into a physical safe set.
3) **Mechanism heterogeneity is flattened.** Converter synchronization, rotor-angle separation, ride-through trip, and limit-cycle boundaries are often evaluated with distinct tools but merged into one TSA output.
4) **Adaptation is not certificate-aware.** Data-efficient transfer selects informative data, but normally does not prioritize the trajectories that make the deployed controller or certificate least defensible.

These are not criticisms of individual papers. They arise because the papers solve different necessary subproblems. The research opportunity lies in integrating them while preserving each method's evidentiary limits.

VI. Combination Gap and Generated Hypotheses

A. Combination Gap

No direct synthesis in the supplied corpus simultaneously links (i) per-event credibility and false-safe risk, (ii) a mode-dependent hybrid stability boundary, (iii) a certificate-filtered operational action, and (iv) label acquisition driven by future certificate risk. Pairwise combinations exist: uncertainty with preventive control [3], credibility with OOD assessment [4], and certificate-based control [7]. The four-way linkage is the combination gap. It is scientifically useful because a downstream action can be safe only if the uncertainty, physical boundary, and deployment update refer to the same operating state and uncertainty assumptions.

B. Hypothesis H1: Credibility-Calibrated Hybrid Stability Certificate

H1. *For renewable-rich systems with topology, fault, and IBR-control shifts, a TSA policy that conditions a mode-specific Lyapunov/barrier certificate on calibrated credibility will yield a lower false-safe rate than a nominal certificate, a pure TSA threshold, and credibility-only abstention at matched operational cost.*

The causal proposal is: PMU/trajectory features and operating context $\rightarrow$ predictive distribution plus credibility $\rightarrow$ dominant instability mode m and robust uncertainty radius u $\rightarrow$ certificate-filtered control set $\mathcal{S}_{\mathrm{rob}}$ $\rightarrow$ realized post-fault security. The phrase "at matched operational cost" is critical: a method should not appear safer merely by curtailing all controllable resources.

H1 has clear counterevidence. It is rejected or narrowed if its robust certificate does not reduce false-safe events, if benefit disappears after cost matching, if calibration deteriorates under unseen mode transitions, or if the mode selector causes more boundary errors than the nominal safe policy.

C. Hypothesis H2: Certificate-Violation-Risk-Driven Active Transfer

H2. *When operating conditions drift, selecting EMT labels by a joint score of OOD distance, predicted misclassification risk, and certificate-violation residual will reach a target false-safe rate with fewer newly labeled trajectories than random, entropy, margin, or OOD-only active transfer.*

For candidate event z, a concrete acquisition score is

$$A(z) = \alpha d_{\mathrm{OOD}}(z) + \beta p_{\mathrm{err}}(z) + \gamma r_{\mathrm{cert}}(z) + \eta q_{\mathrm{mode}}(z), \quad (8)$$

where d_{OOD} denotes distributional novelty, p_{err} predicted classification error, r_{cert} a barrier/Lyapunov residual or near-violation score, and q_{mode} rewards underrepresented mechanism modes. Coefficients must be tuned only on training/validation conditions; otherwise the claimed labeling efficiency may leak target-test knowledge.

H2 is falsified if its lower label count is achieved only through more conservative operation, if it forgets previously reliable regimes, or if its safety rate is no better than simpler acquisition baselines at equivalent EMT budget.

VII. Feasible Experimental Programme and Expected Results

A. Testbed and Data Protocol

Use three escalating testbeds: IEEE 39-bus and IEEE 118-bus systems for comparability with current PTSA studies [1]; a renewable-rich switched-system case with LVRT/FRT trip logic [8]; and a multi-VSC network with PLL, current-transient, nonlinear-damping, and interaction effects [9], [10]. Generate RMS trajectories for broad coverage and reserve EMT simulations as the authoritative label source near boundaries. A subset should be evaluated by RT-LAB/HIL when available, because HIL provides an empirical check on the model abstraction for multi-VSC transients [10].

Partition data by *scenario*, not random trajectory: hold out fault locations, clearing times, renewable penetration ranges, line outages, PLL parameters, control modes, and topologies. This prevents the misleading result in which nearly identical trajectories appear in both training and testing. Record a mechanism label from simulation diagnostics: rotor separation, voltage violation, PLL loss of synchronization, protection/trip cascade, or limit-cycle proximity.

B. Models, Baselines, and Ablations

The full method contains a trajectory encoder, probabilistic margin/risk head, credibility head, mode selector, mode-specific certificate bank, safety filter, and active-transfer module. Compare against: (B1) deterministic TSA threshold; (B2)

TABLE II
PRIMARY OUTCOMES AND SUCCESS CRITERIA

Outcome	Interpretation
False-safe rate	Fraction of declared-safe events that violate the authoritative EMT/HIL security criterion; primary safety outcome.
Risk calibration	ECE, Brier score, and risk-coverage curves, reported overall and near the stability boundary.
Certificate validity	Frequency/magnitude of barrier or Lyapunov residual violation, separated by identified mode.
Operational impact	Load shedding, renewable curtailment, control effort, and recovery time at a fixed reliability target.
Adaptation efficiency	EMT labels required to reach target false-safe rate and calibration under each held-out shift.

uncertainty-aware TSA without credibility-to-action mapping; (B3) credibility abstention without a certificate; (B4) nominal neural Lyapunov/barrier control; and (B5) active transfer using random, entropy, margin, and OOD-only acquisition.

Remove one component at a time: credibility conditioning, mode-specificity, robust uncertainty inflation, certificate-residual acquisition, and physics auxiliary loss. The most important ablation is H1 without the credibility-to-safe-set link; it reveals whether a credibility score is merely cosmetic.

C. Expected Outcomes and Interpretation

The expected outcome is not perfect prediction. A successful H1 system should move ambiguous cases from unsafe "false safe" decisions to either conservative but feasible actions or escalation, reducing false-safe events with a quantifiable cost trade-off. A successful H2 system should concentrate expensive EMT labels near the intersection of novelty, error risk, and certificate fragility. If only accuracy rises but false-safe rate, calibration, or cost does not improve, the central hypotheses are not supported. This is a valuable outcome because it identifies which bridge in (1) failed.

VIII. Readiness, Risk, and Reproducibility

Research readiness should not be confused with a claim that the hypotheses are true. We distinguish: (i) *corpus readiness*, supplied by the 45-paper corpus and its coverage of all six SH functions; (ii) *evidence readiness*, supplied by direct theoretical, simulation, and HIL-compatible tests; and (iii) *scientific assessment*, which remains "inconclusive" until the planned comparisons are completed. Reproducibility requires versioned network models, fault and control parameter grids, simulator versions, train/validation/test scenario splits, random seeds, calibration code, certificate residual definitions, and all failed cases. Results should report both successful and failed certificate checks rather than silently excluding unstable or OOD events.

IX. Conclusion

The literature corpus supports a focused research programme beyond another high-accuracy TSA classifier. Its strongest unexploited opportunity is the closed loop from credibility to a hybrid physical certificate, from that certificate to an action, and from operational failure risk back to high-value labels. H1 and H2 supply distinct but composable contributions: H1 makes online decisions more defensible; H2 makes continued adaptation more data-efficient without treating label count as the final objective. Their value is conditional and falsifiable through false-safe rate, calibration, certificate validity, and cost-aware testing across genuinely unseen operating regimes.

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